PYTHON FOR CYBERSECURITY ROADMAP

A comprehensive 17-phase technical guide to mastering Python for offensive security, threat detection, forensics, and tool development.

PHASE 1 Python Fundamentals

1. Introduction to Python
2. Installation & Environment Setup
3. Python Interpreter & Syntax
4. Variables & Data Types
5. Type Casting & Operators
6. Conditional Statements
7. Loops (For & While)
8. Break, Continue and Pass
9. Functions & Lambda Functions
10. User Input and Output
11. Comments and Documentation

PHASE 2 Python Data Structures

1. Strings, Lists, Tuples
2. Sets & Dictionaries
3. List & String Methods
4. Dictionary Methods
5. List Comprehensions
6. Nested Data Structures
7. Slicing Concepts
8. Iterators & Generators

PHASE 3 Advanced Python Fundamentals

1. Modules, Packages & Import System
2. Exception & File Handling
3. JSON & CSV Handling
4. Regular Expressions (Regex)
5. Object-Oriented Programming (OOP)
6. Classes, Objects & Inheritance
7. Encapsulation & Polymorphism
8. Decorators & Context Managers

PHASE 4 Linux & Python Integration

1. Python in Linux Environments
2. Virtual Environments & Package Mgmt
3. Command-Line Arguments & Argparse
4. Subprocess, OS, Sys & Pathlib Modules
5. Environment Variables
6. Linux File & Process Management
7. Shell Command Execution

PHASE 5 Networking with Python

1. Networking Fundamentals
2. Socket Programming (TCP/UDP Clients & Servers)
3. Port Scanning & Banner Grabbing
4. IP Handling & DNS Resolution
5. Host Discovery & Network Connections
6. Packet Capture & Packet Analysis
7. Scapy Fundamentals & Traffic Monitoring

PHASE 6 Python for Information Gathering

1. OSINT Fundamentals & HTTP Requests
2. URL Parsing & DNS/WHOIS Lookup Auto
3. IP Geolocation & Subdomain Enumeration
4. Port Scanning Auto & Service Detection
5. Metadata & Email Header Analysis
6. API Data Collection & Recon Automation

PHASE 7 Python for System Security

1. System Info Gathering & User Enum
2. Process & Service Monitoring
3. File Permission & Integrity Auditing
4. Hash Generation & Password Strength Check
5. Log File Analysis & Resource Monitoring
6. System Hardening Automation

PHASE 8 Python for Cryptography

1. Cryptography Fundamentals
2. Hashing Algorithms (MD5, SHA-1/256/512, HMAC)
3. Symmetric Encryption (AES)
4. Asymmetric Encryption (RSA, Digital Signatures)
5. Base64 Encoding & Decoding
6. Password Hashing & Key Management

PHASE 9 Python for Web Security

1. HTTP Requests/Responses & Requests/BS4
2. URL, Header, Cookie & Session Analysis
3. API Security & Auth Testing
4. Security Headers & SSL/TLS Cert Analysis
5. Web Tech & Misconfiguration Detection

PHASE 10 Python for Vulnerability Assessment

1. Port Scanning & Service Enumeration Auto
2. Directory & Subdomain Enum Automation
3. DNS & SSL Misconfiguration Detection
4. Custom Vulnerability Scanner Development
5. CVE Data Collection & CVSS Score Analysis

PHASE 11 Python for SOC & Blue Team

1. Log Collection, Parsing & Analysis (Win/Linux)
2. Failed Login & Brute-Force Detection
3. IP/Hash Reputation & Threat Intel API Integration
4. File Integrity Monitoring & Alert Generation
5. Incident Response Automation & SIEM Integration

PHASE 12 Python for Malware Analysis

1. PE File & Metadata Analysis
2. String Extraction & Entropy Analysis
3. Static/Dynamic Malware Analysis & Sandbox Auto
4. Malware IOC Extraction & YARA Integration
5. Malware Classification & Reporting

PHASE 13 Python for Digital Forensics

1. File System & Metadata Analysis
2. Hash Verification & Timeline Analysis
3. Browser Artifact & Event Log Forensics
4. Memory & Network Forensics Basics
5. Evidence Collection Automation & Reporting

PHASE 14 Security Libraries & Frameworks

Requests **Scapy** **Socket** **Paramiko** **Nmap**
BeautifulSoup **Cryptography** **PyCryptodome** **Psutil**
Watchdog **YARA-Python** **Volatility** **Impacket** **PyShark**
Rich **Colorama**

PHASE 15 Security Tool Development

1. CLI Security Tool Development
2. Network & Port Scanner Development
3. Subdomain & DNS Enumeration Tools
4. WHOIS Lookup & IP Reputation Checkers
5. Password Strength & Hash Tools
6. File Integrity Monitor & Log Analyzer
7. Packet & Vulnerability Scanners
8. IOC Extractor & Malware Tools

PHASE 16 Advanced Python Concepts

1. Multithreading & Multiprocessing
2. Asynchronous Programming (Asyncio)
3. API & Database Integration (SQLite)
4. Regex Optimization & Memory Mgmt
5. Performance & Secure Coding Practices
6. Error Logging, Debugging & Unit Testing
7. Automation Framework Development

PHASE 17 Practical Hands-on Projects

1. Automated Network Scanner
2. Multi-Threaded Port Scanner
3. Subdomain Enumeration Tool
4. DNS Reconnaissance Tool
5. Network Packet Sniffer
6. Password Security Auditor
7. File Hashing Utility
8. File Integrity Monitoring System
9. Linux Log Analyzer
10. Windows Event Log Analyzer
11. Brute-Force Detection System
12. IP Reputation Checker
13. Threat Intelligence Aggregator
14. Basic Malware Analysis Tool
15. YARA-Based File Scanner
16. Automated Vulnerability Assessment Tool
17. SOC Alert Automation System
18. Python-Based Security Toolkit